

SRN: PES1UG24AM380

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Problem Statement #64 — Disaster Relief Supply & Volunteer Coordinator

Actors: Relief Volunteer, Disaster Manager **Secondary/System Actor:** Warehouse Inventory System

1. Functional Requirements

ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall match incoming field shelter supply requisitions against available central warehouse supplies and allocate delivery priority.	High	Pass: Critical need given highest dispatch priority and routed to logistics manifest. Fail: Low-priority item dispatched ahead of water/medical needs.	Ensures scarce relief supplies reach life-critical needs first, preventing triage errors during high-demand periods.
FR-002	The system shall dispatch volunteers to shelters by matching each volunteer's registered skill set (medical, logistics, translation, etc.) to the skill requirement of the open task.	High	Pass: A volunteer with a "Medical" skill tag is assigned before a general volunteer when a medical task is open. Fail: An unskilled volunteer is assigned to a task requiring certification.	Misassigned volunteers slow relief operations and can create safety risks in specialized tasks (e.g., first aid).
FR-003	The system shall track the real-time location and delivery status (En Route, Delivered, Delayed) of each relief vehicle against its assigned manifest.	Medium	Pass: Manager dashboard reflects a vehicle's status change within 60 seconds of a field update. Fail: Vehicle status remains stale after a field update is submitted.	Managers need live visibility to re-route vehicles around blocked roads or reprioritize deliveries.
FR-004	The system shall allow a Relief Volunteer to register and update a profile containing skills, availability windows, and certifications.	Medium	Pass: Updated availability is reflected immediately in the volunteer-matching pool. Fail: A volunteer marked "unavailable" is still offered new dispatch assignments.	Accurate, self-maintained volunteer data is required for FR-002 to produce valid matches.
FR-005	The system shall automatically generate a low-stock alert to the Disaster Manager when any critical supply category (water, blankets, medical kits) falls below a configurable threshold at a shelter.	High	Pass: Alert is raised and visible on the dashboard within 5 minutes of stock crossing the threshold. Fail: Stock depletes to zero with no alert raised.	Proactive alerts prevent shelters from running out of critical supplies before a scheduled resupply.

2. Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The disaster mapping dashboard must operate with low network bandwidth consumption and support offline GIS data caching.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Dashboard fails to load cached map data when connectivity drops.	Disaster zones frequently have degraded or intermittent connectivity, so the system must remain functional offline.
NFR-002	Usability & Reliability	The system must be operable by volunteers with minimal (under 15 minutes) training and maintain at least 99.5% uptime during an active disaster response window.	High	Pass: A first-time volunteer completes a requisition submission unassisted within 15 minutes of onboarding; system logs show uptime ≥ 99.5% during a simulated 72-hour incident. Fail: Volunteers require developer/admin support to complete basic tasks, or the system experiences unplanned downtime during the incident window.	Field volunteers are often untrained in software use and under time pressure; the platform cannot be a bottleneck during a live emergency.

3. Use-Case Flow Specification

Use Case: Allocate Delivery Priority (FR-001)

Actor(s): Disaster Manager (primary), Warehouse Inventory System (<<include>>)

Preconditions:

- The Disaster Manager is authenticated and logged into the coordination dashboard.
- At least one open shelter supply requisition exists in the system.
- The Warehouse Inventory System has current stock records available.

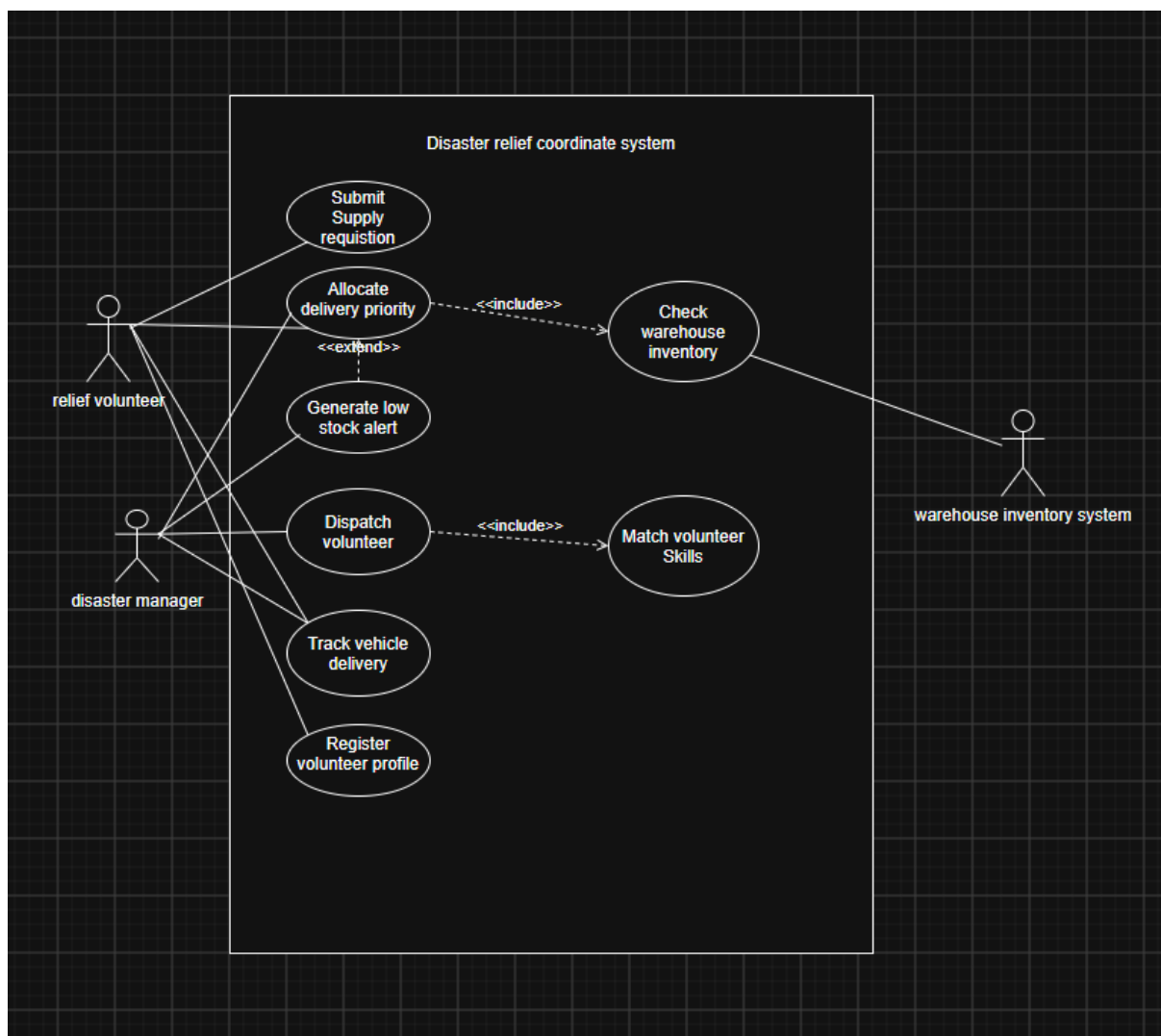
Postconditions:

- The requisition is assigned a delivery priority level (Critical / High / Standard).
- The prioritized requisition is added to the logistics dispatch manifest.
- If stock is insufficient, the requisition is flagged as "Pending Resupply" instead of being dropped.

Main Success Scenario:

1. A Relief Volunteer submits a shelter supply requisition (item type, quantity, shelter ID).

2. The system receives the requisition and triggers the Check Warehouse Inventory use case (<<include>>) to confirm stock availability.
3. The system classifies the requisition's urgency based on item type (e.g., water/medical kits are auto-flagged Critical).
4. The system checks existing manifest load and available delivery capacity.
5. The system assigns a priority level and inserts the requisition into the logistics manifest in priority order.
6. The Disaster Manager reviews the updated manifest on the dashboard and confirms dispatch.
7. The system notifies the assigned relief vehicle/driver of the new manifest entry.



Alternate Flow — Insufficient Warehouse Stock (A1):

1. At step 2, the Check Warehouse Inventory use case returns insufficient stock for the requested item.
2. The system marks the requisition as "Pending Resupply" rather than assigning it to the manifest.

3. The system triggers the Generate Low-Stock Alert use case (<<extend>> on this use case, extension point: *Stock Below Threshold*) to notify the Disaster Manager.
4. The Disaster Manager may manually override and approve partial fulfillment, or wait for resupply.
5. Flow resumes at step 6 once stock becomes available or a manual override is issued.